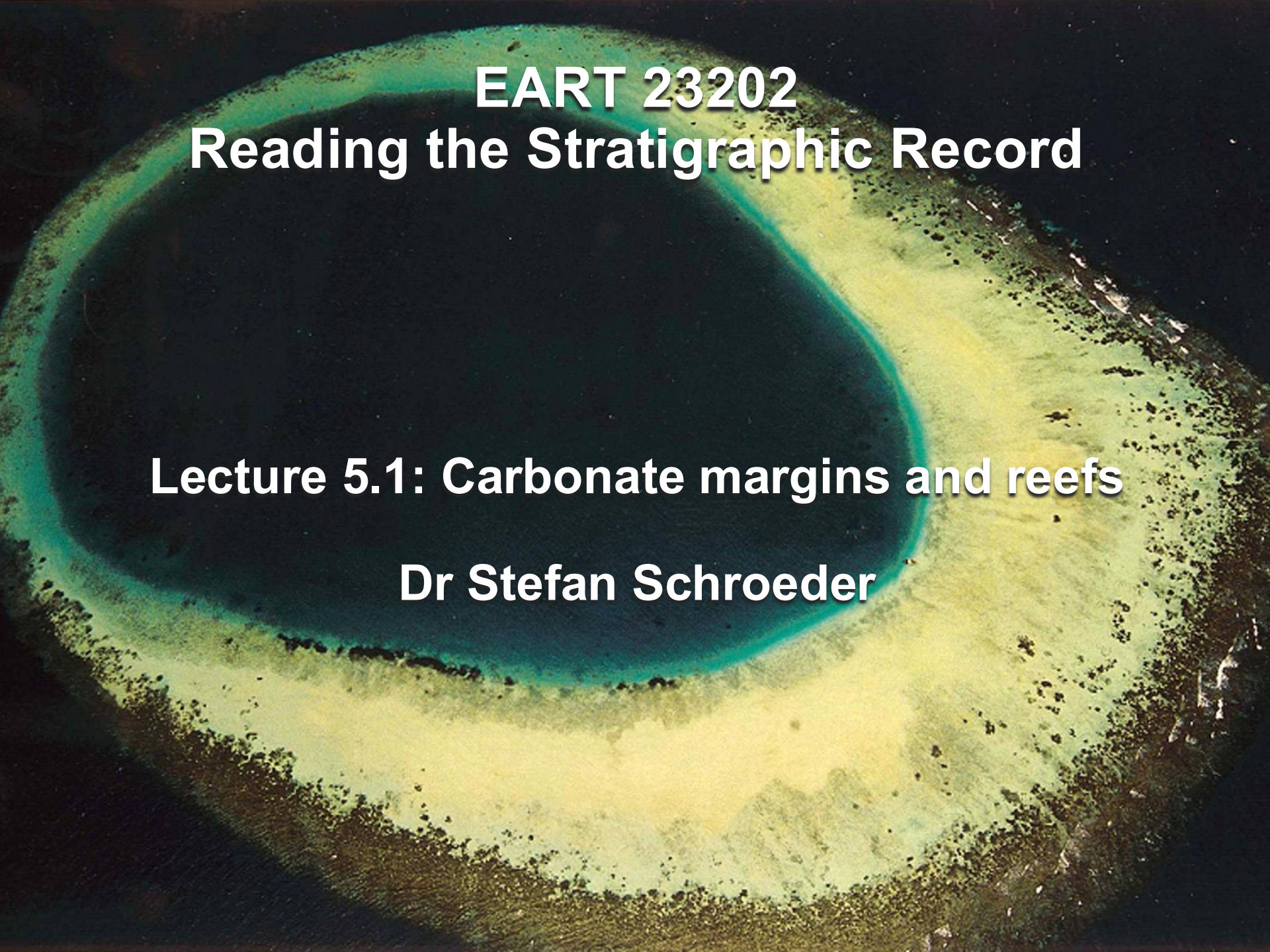


EART 23202
Reading the Stratigraphic Record

Lecture 5.1: Carbonate margins and reefs

Dr Stefan Schroeder

Carbonate reefs and build-ups
Definition

Carbonate build-ups are laterally-restricted structures, which have usually undergone organically-mediated growth. They can be grossly divided into:

Organic (skeletal) reefs, built by organisms with a rigid calcareous frame, may be matrix or skeleton supported and deposited in warm or cold water and able to withstand high energy wind/wave action

Reef (mud) mounds are inorganically and/or biogenically constructed but lack a rigid skeletal framework and unable to withstand high energy wind/wave action

Photozoa
T Factory

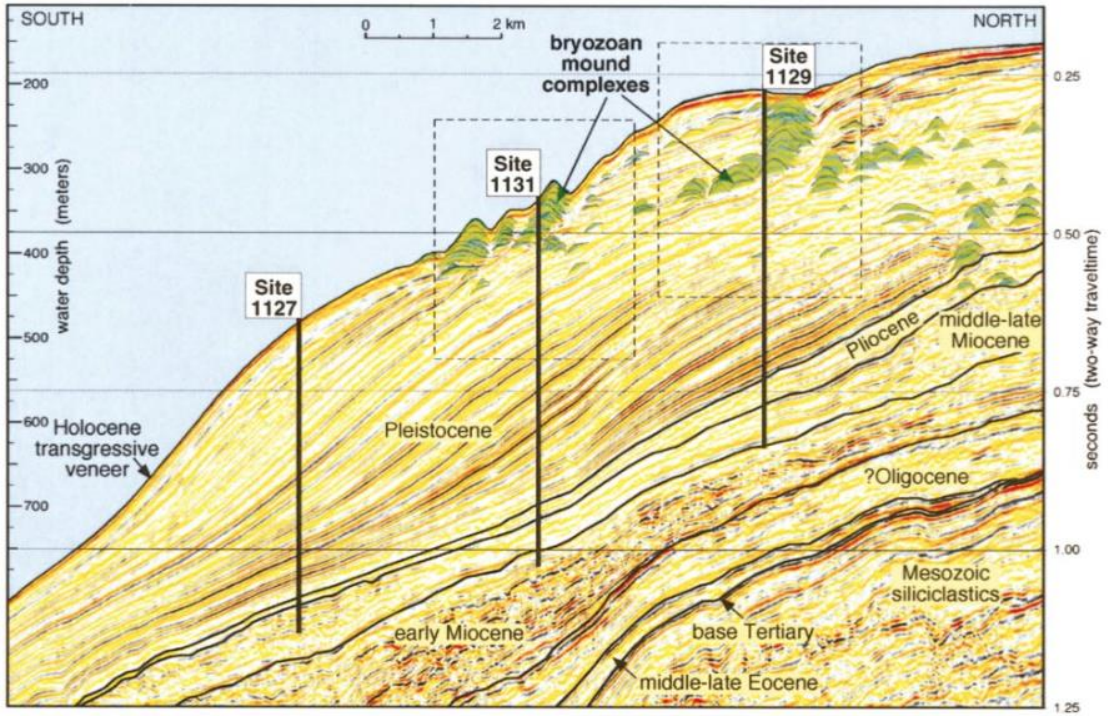
Heterozoa
C Factory

Microbes
M Factory

Carbonate reefs and build-ups
Definition



Great Barrier Reef (NASA)



Australia (James 2004)

Photozoa
T Factory

Heterozoa
C Factory

Carbonate reefs and build-ups
Reef and mud mounds



http://www.uni-graz.at/bernd.kaufmann/mudmounds.htm



- **'Fine grained, mud (micrite)-dominated build-ups**
- **Stability provided by matrix, limited cementation**
- **Organic components include bivalves, corals, sponges, bryozoa, microbes, stromatoporoids**
- **Heterotrophic and biologically influenced/induced carbonate precipitation**
- **Low topographic relief - do not offer significant wave resistance, but can trap sediment**
- **When skeletal framebuilders were absent or in deeper-water setting (common Paleozoic)**

Walker, 1984

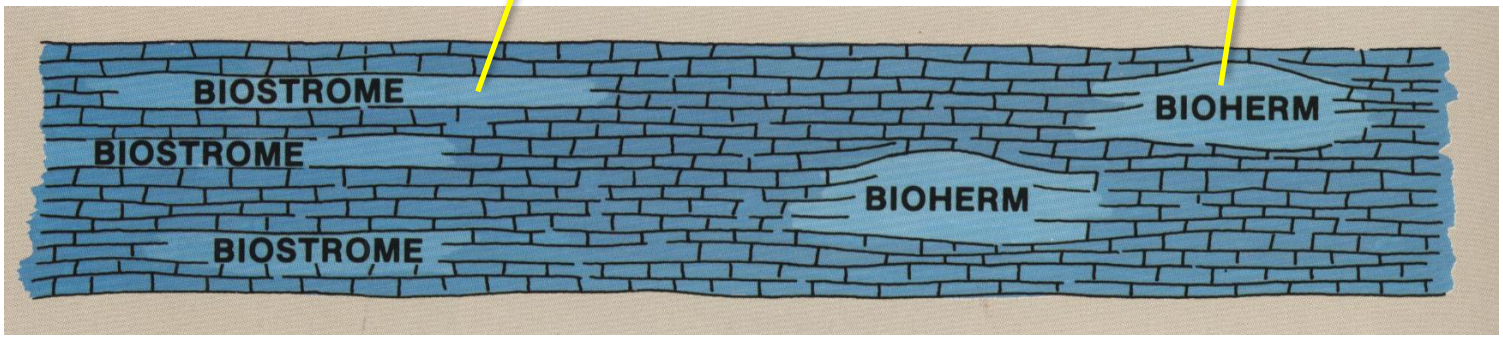


Carbonate reefs and build-ups
Biostrome vs bioherm



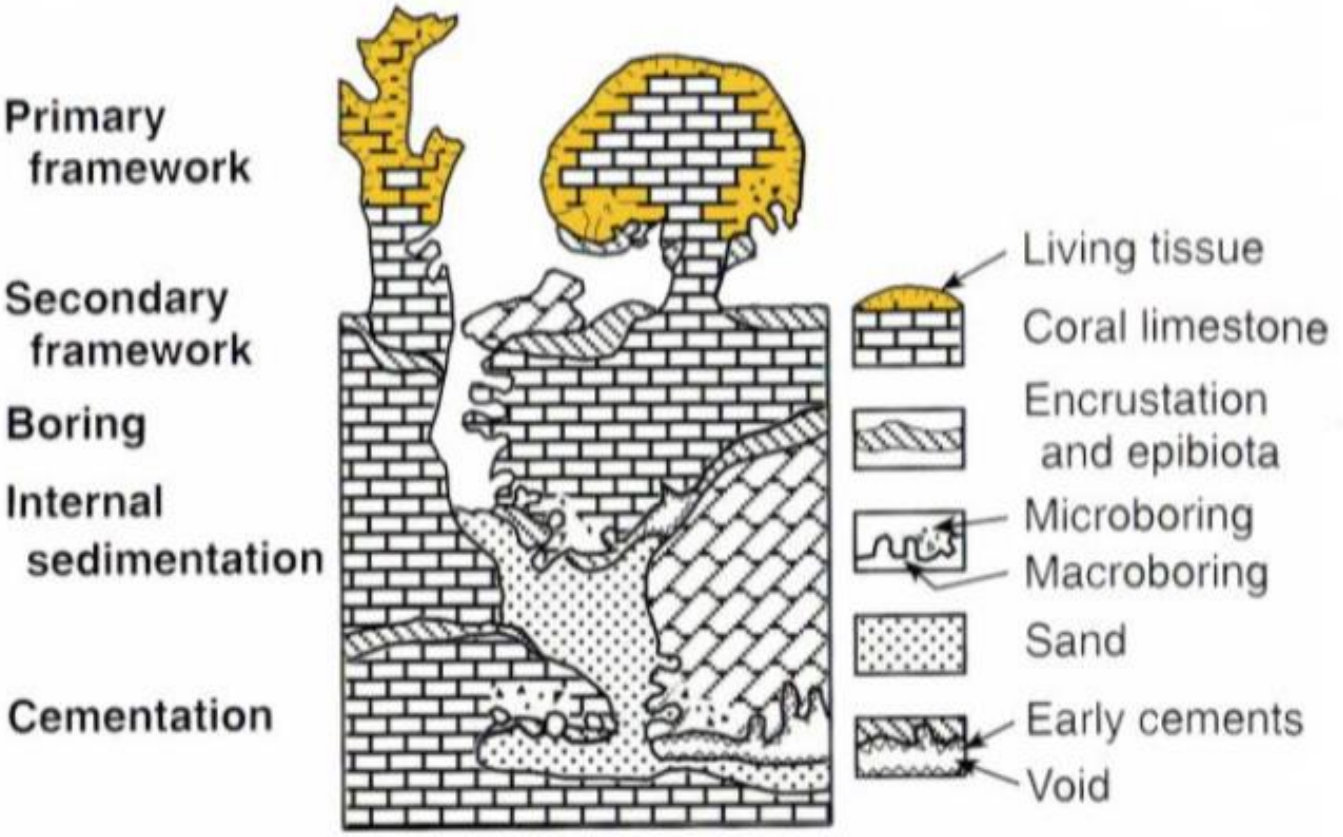
Jurassic, Morocco





Scholle (1983)

Reef dynamics and ecology



Woodroffe, 2003

<u>Constructive processes</u>:
Biological processes through direct growth, baffling or binding

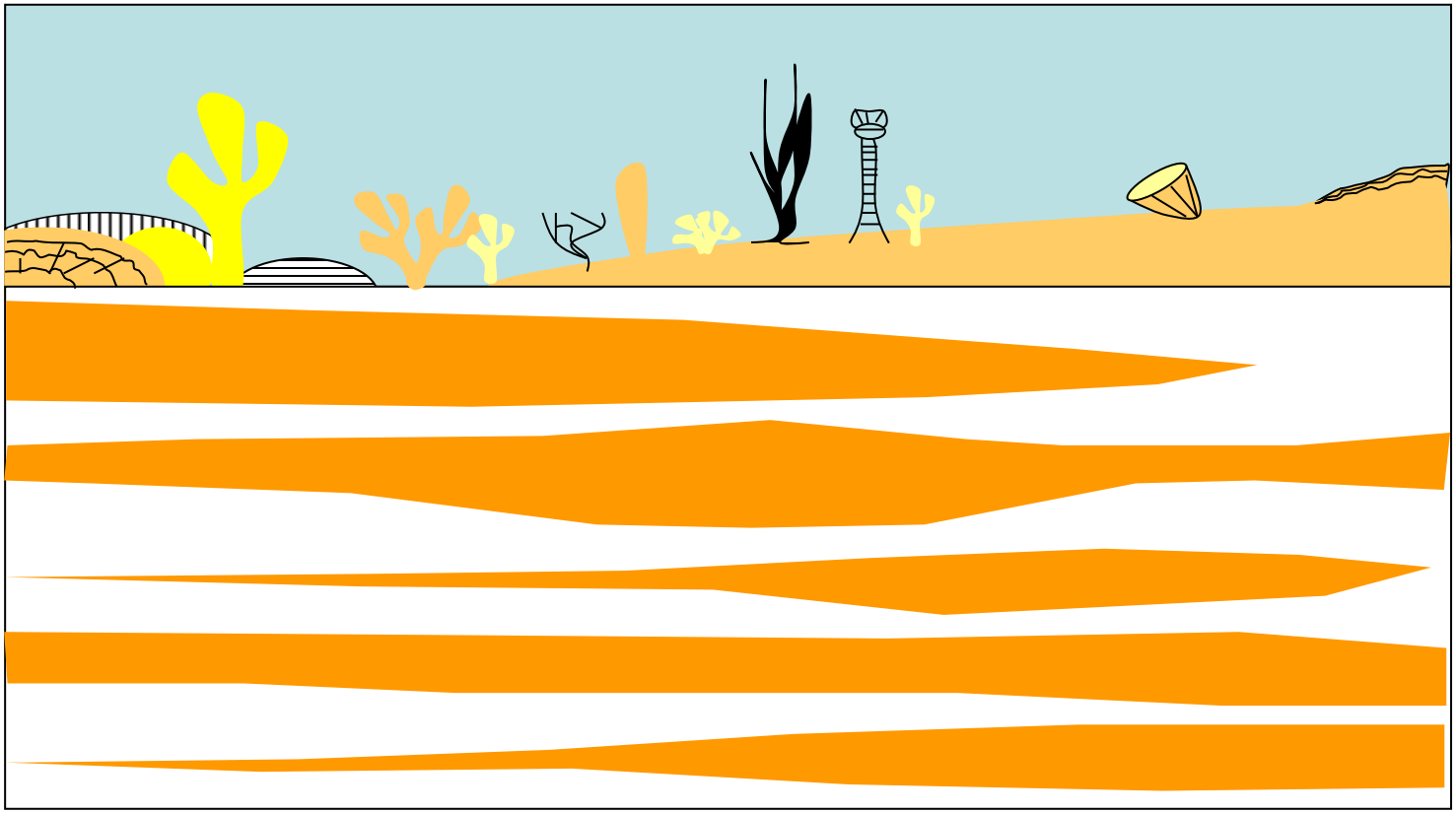
<u>Destructive processes</u>
Wave damage and biological destruction

<u>Cementation</u>
Early cementation from seawater

<u>Sedimentation</u>
Accumulation of biogenic matter and reef-derived detritus

Reef dynamics and ecology
Organism roles

Frame-built	Reef mound		Mud mound
Coral Stromatoporoid Red algae Stromatolites	Bryozoan Phylloid algae Sponges	Codiacean algae Sea grass Crinoids	Microbial mats



Frame builders

Sediment contributors

Bafflers

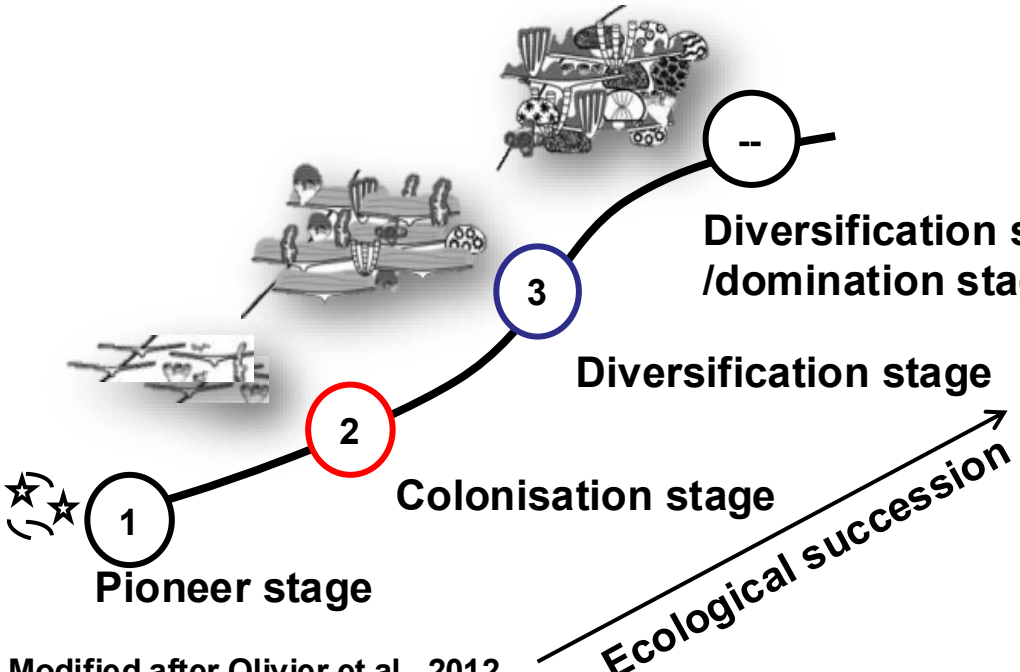
Binders

Precipitators

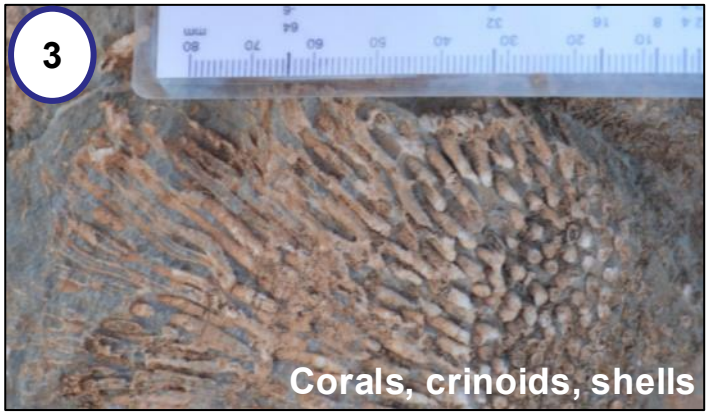
Modified from Tucker and Wright, 1990

Reef dynamics and ecology
Biological succession in reefs

Oxfordian reef ecological succession (Morocco)



Modified after Olivier et al., 2012

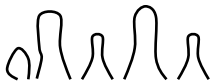
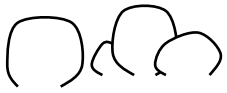
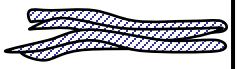
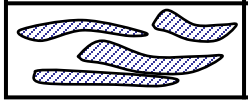




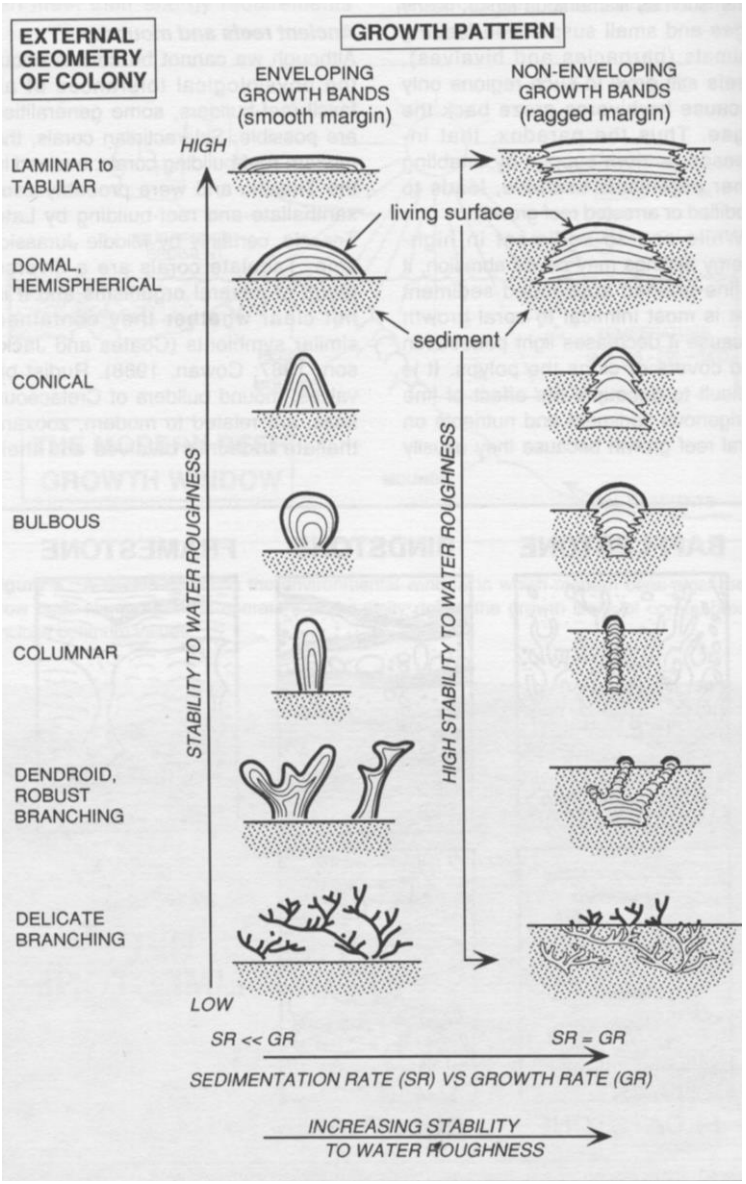




Reef dynamics and ecology
Growth form of frame-building organisms

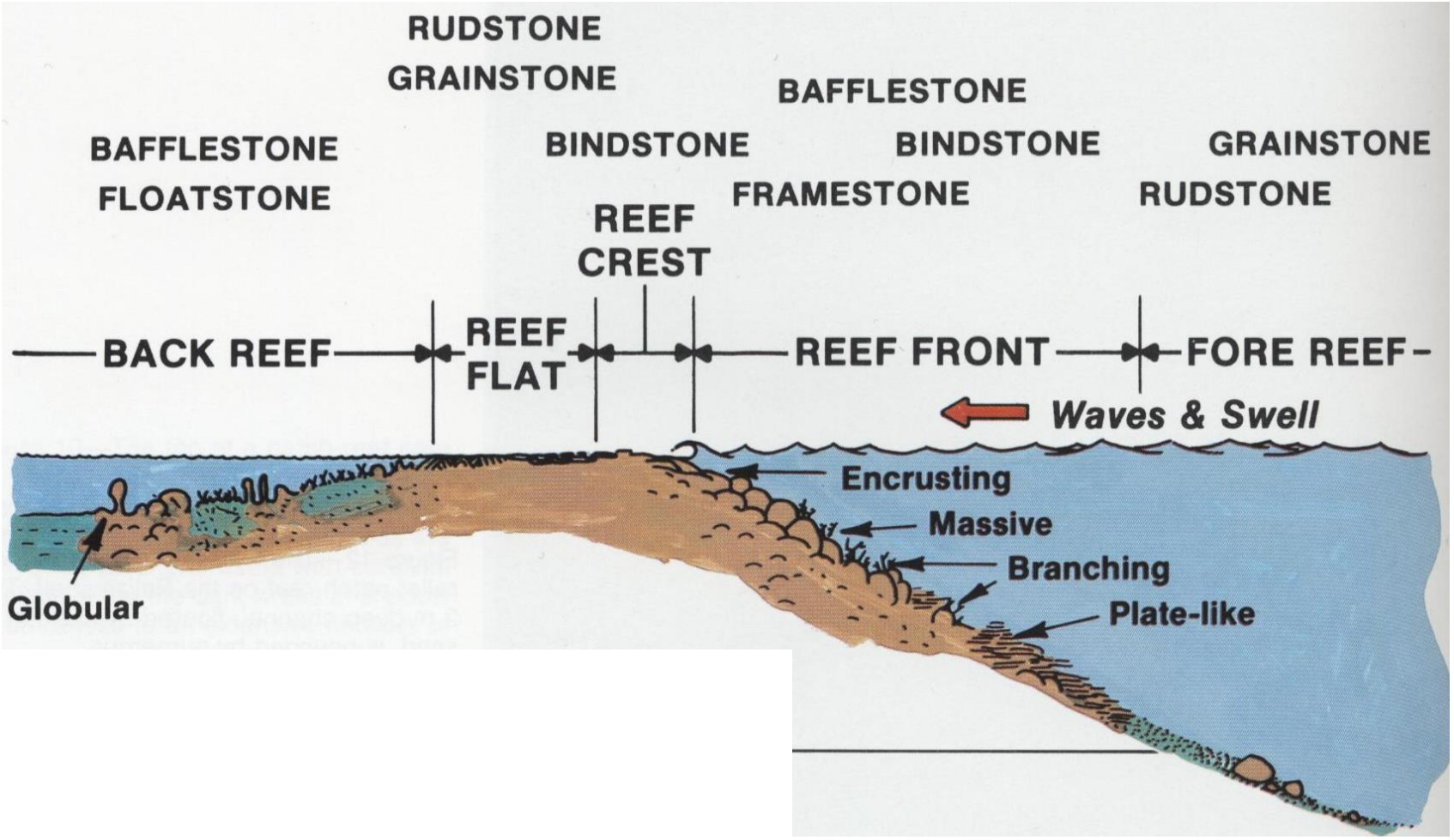
Growth form		Wave energy	Sedimen-tation
	Delicate, branching	Low	High
	Thin, delicate, plate-like	Low	Low
	Globular, bulbous, columnar	Moderate	High
	Robust, dendroid, branching	Moderate - high	Moderate
	Hemispherical, domal, irregular, massive	Moderate - high	Low
	Encrusting	Intense	Low
	Tabular	Moderate	Low

Modified from Pomar et al, 1985; Tucker and Wright, 1990



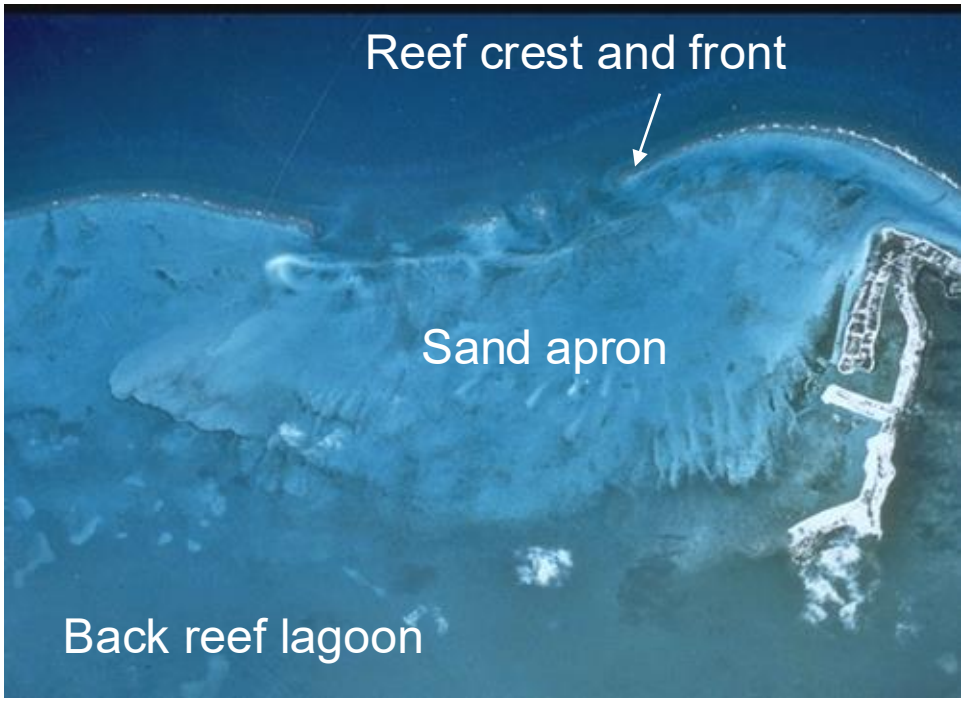
Walker (1984)

Reef facies



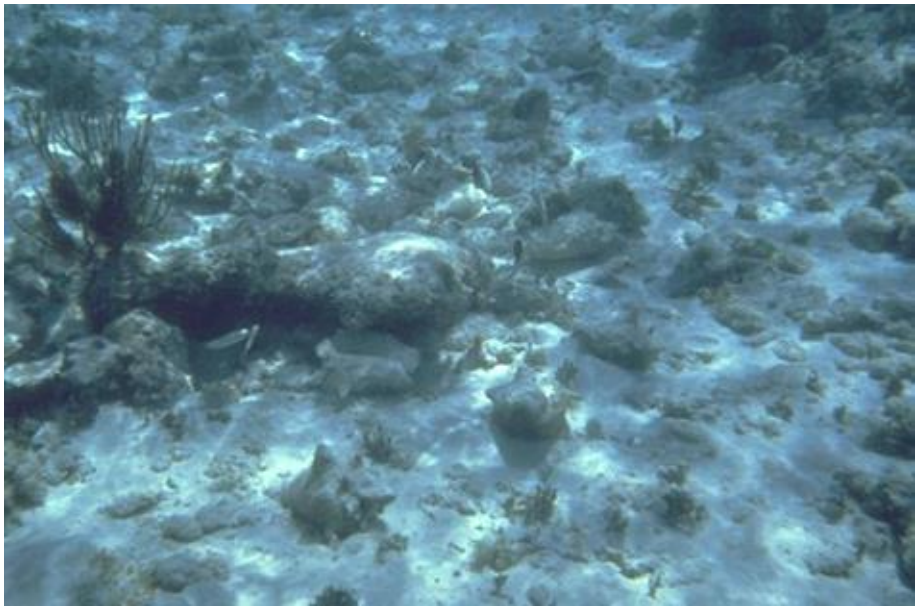
After James (1979) in Scholle (1983)

Reef facies
Reef flat: sand apron



Harry Roberts/SEPM/ Grand Cayman

J. Mazzullo/SEPM



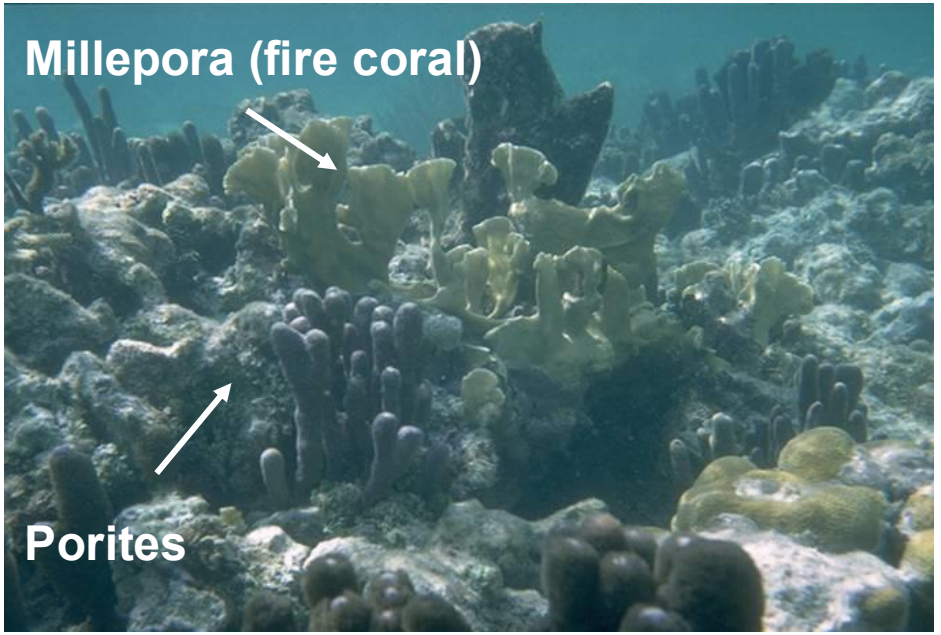
Landward edge of reef flat with large gastropods (Strombus gigas), soft corals, and *Halimeda* on gravelly substratum. L.A. = 1.7 m

- **Behind reef pavement, water depths up to 10m. May be up to 160km long and 100-200m wide**
- **Reworked reef debris, carbonate sand, local colonisation by sea grass and algal mats**
- **Some coral growth, intense bioerosion, algal encrustation of boulders**
- **Gradational contact with back-reef lagoon**

Reef facies
Reef crest: composition

- Dominated by encrusting organisms, especially red algae, usually coating dead coral/coral debris. May be encrusted by forams, gastropods etc
- Low energy crests may be composed of *Millepora* or *Acropora Palmata*
- Skeletal breakage, abrasion, bioerosion high
- Periodic subaerial exposure possible
- Bindstones/framestones in ancient carbonates, with laminar encrusting organisms



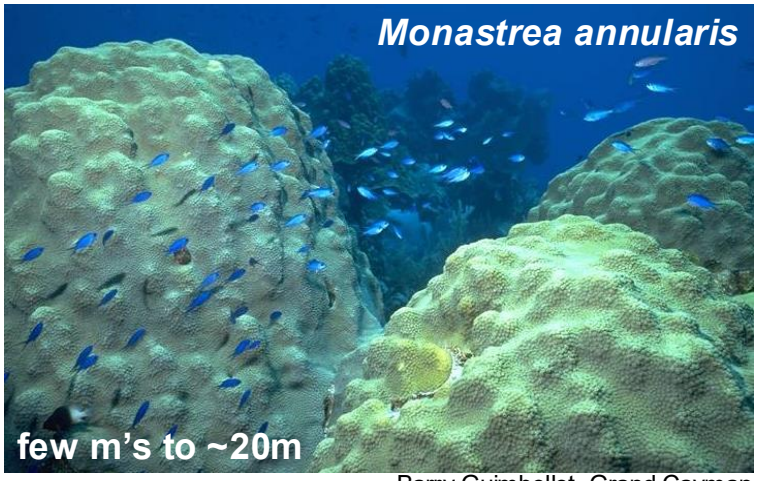


Ed Purdy/SEPM

Reef facies
Reef front

- Extensive coral growth seaward of reef crest: ‘reef core’ preserved in ancient reef limestones
- Close to the crest, in the high energy zone, spur and groove structures form oblique to the shoreline
- Biota evolves with depth as light penetration and energy decrease
- Low preservation potential due to bioerosion and early diagenesis

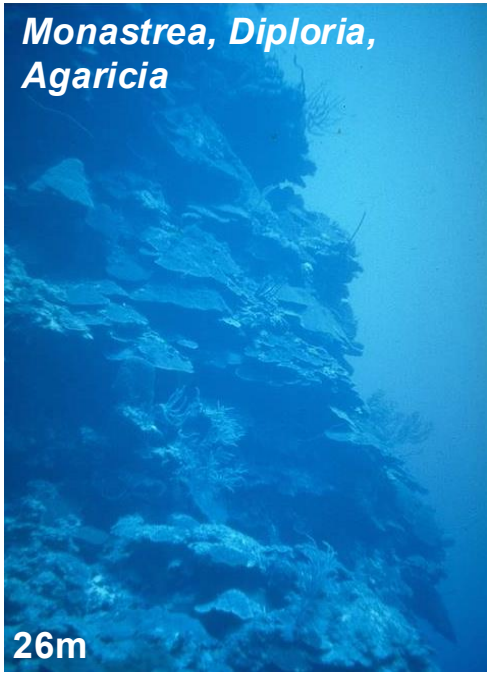


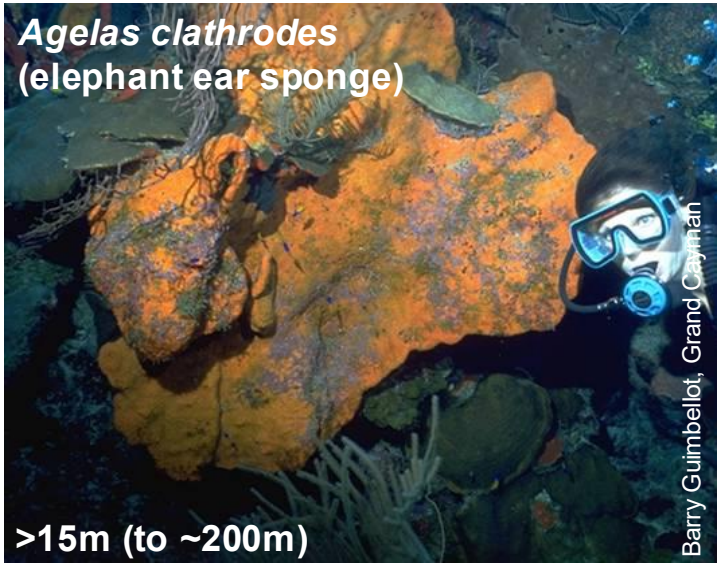


Barry Guimbellot, Grand Cayman



W.Mayhew (1988)/Quintana Roo, Mexico

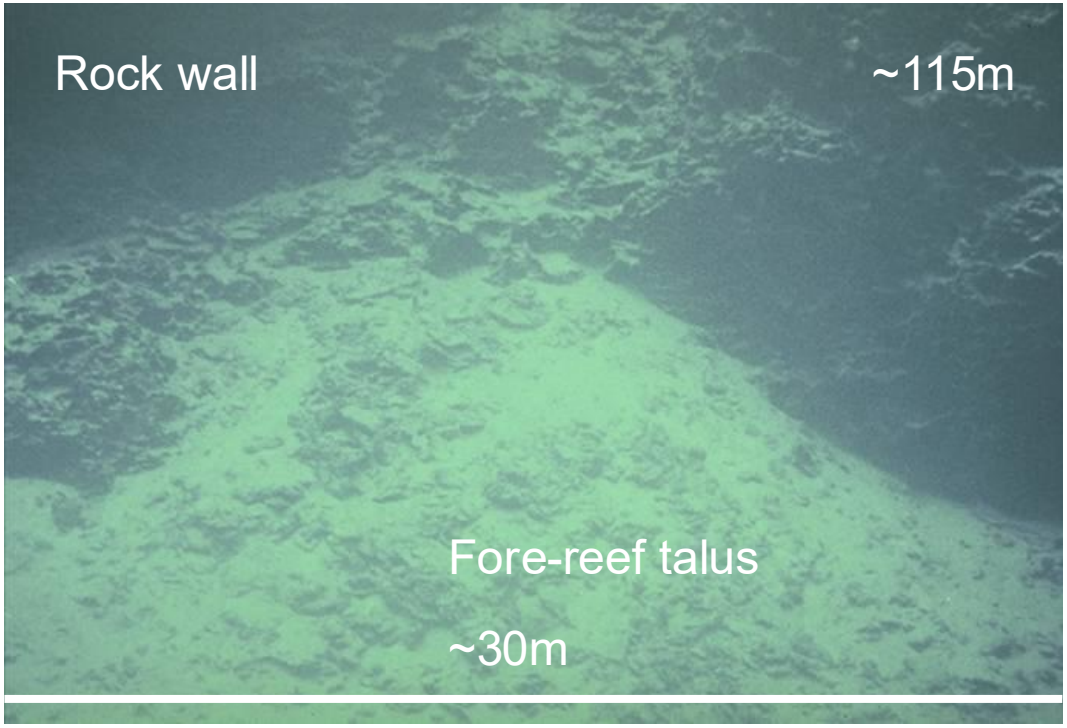




Reef facies
Forereef slope



The coarse fore-reef sediment rich in *Halimeda* plates from a water depth of 1025 meters.
Scale bar is 2 mm long



Noel P. James/SEPM/ Belize Barrier

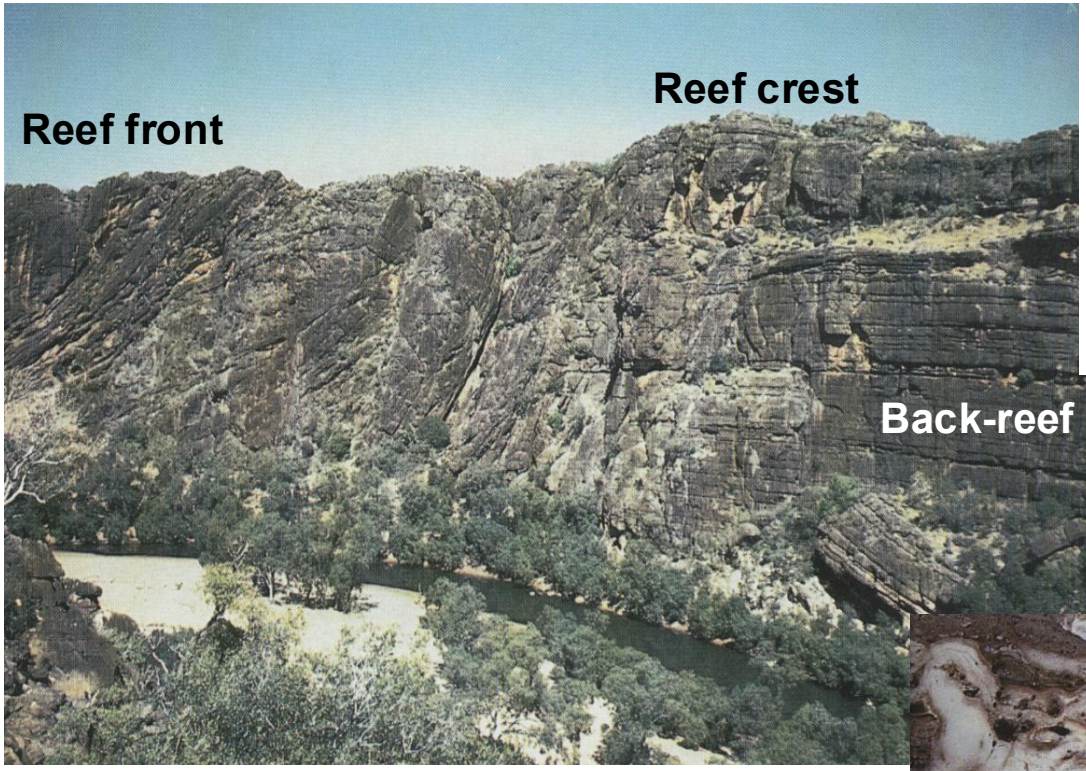
- **Forereef slope is positioned seaward of the reef front, transition to basin**
- **Sedimentation dominated by gravity flow mechanisms and deposition of pelagic sediments**
- **Depositional/accretionary reef margins slope continuously into the basin**
- **By-pass margins have a steep escarpment seperating the reef from reef talus**

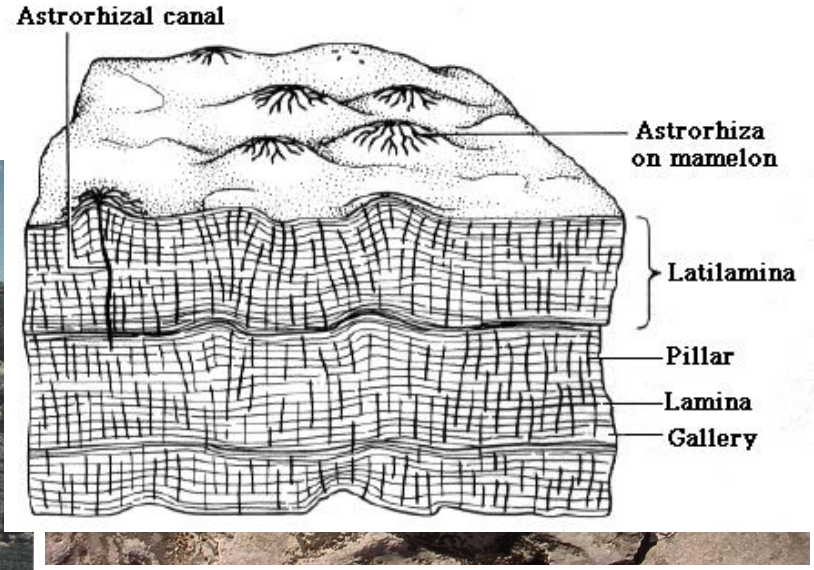
Ancient reefs and buildups



Ancient reefs

Devonian reef complex, Australia



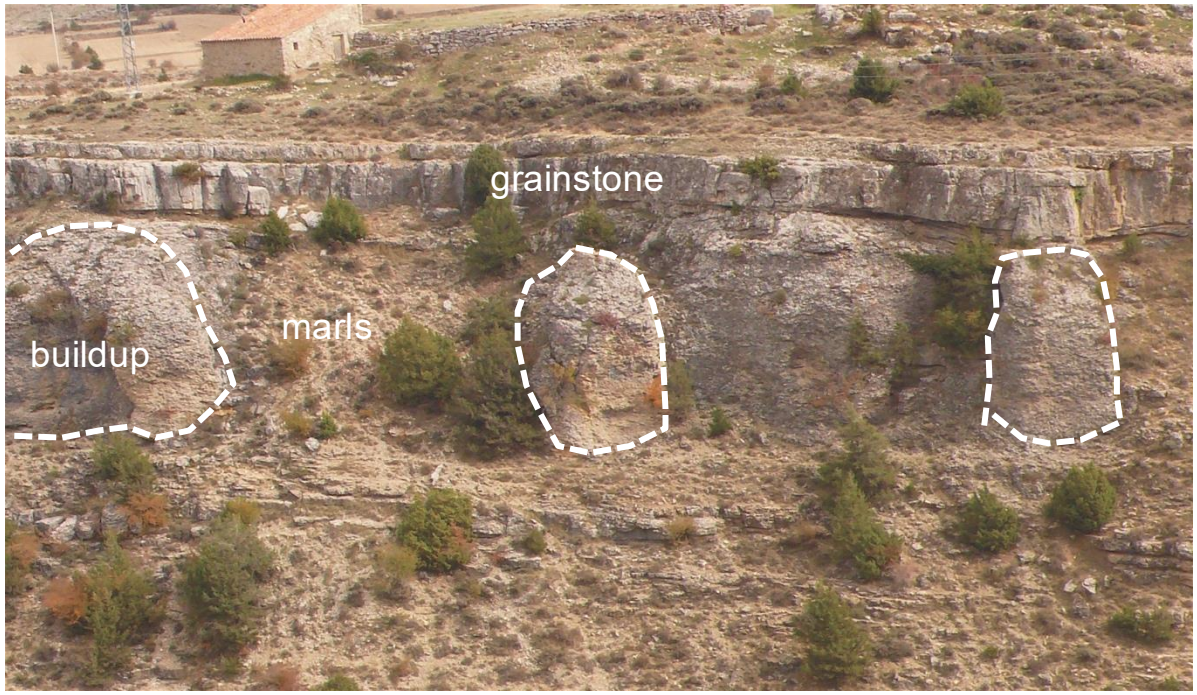


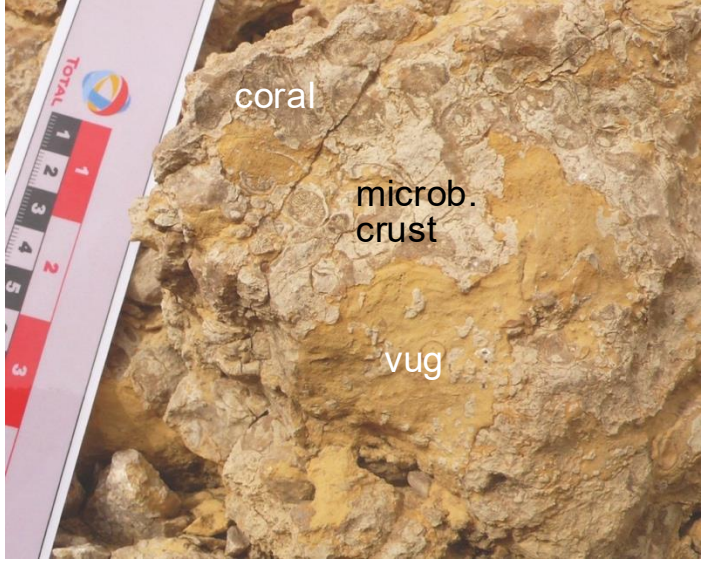


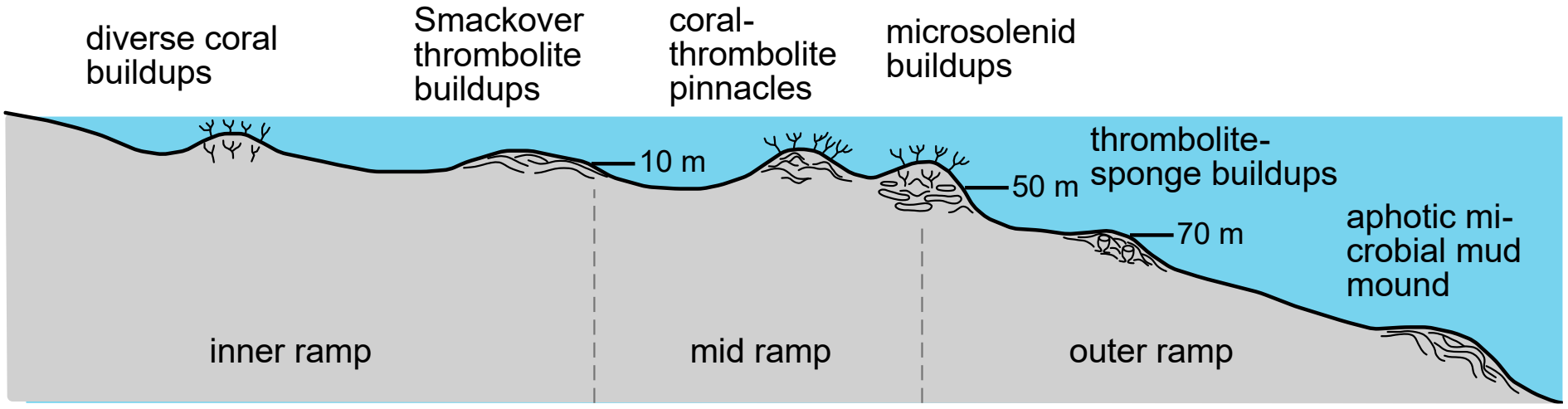
- Small reef mounds to barrier reefs
- Framebuilders: stromatoporoids, corals, plus sponges, bryozoa, algae, echinoderms

Ancient reefs

Jurassic coral-microbe bioherms







Mancini, 2008

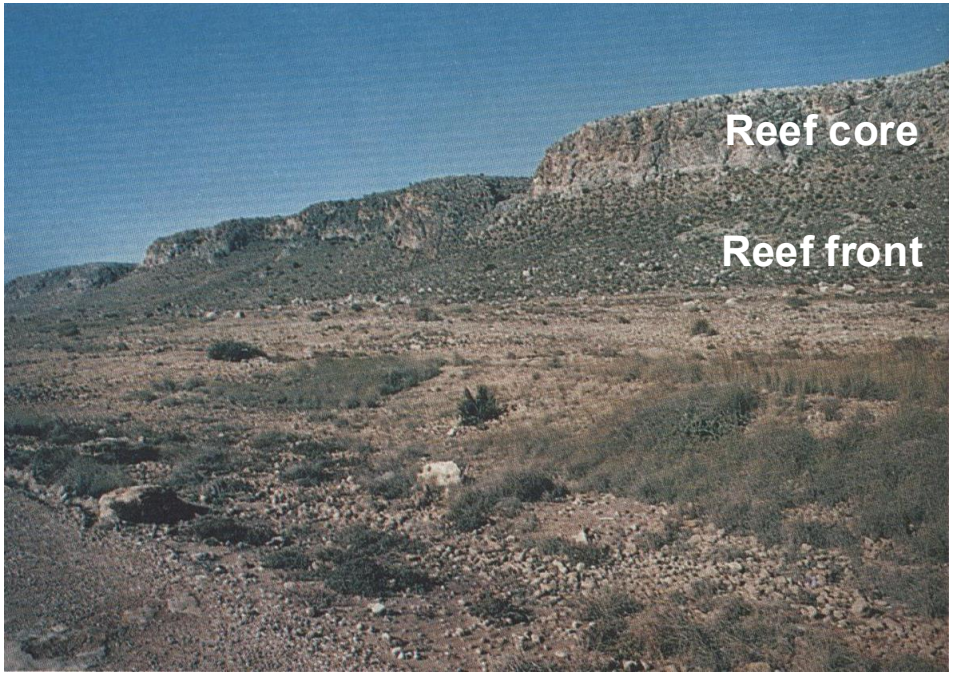
Ancient reefs
Cretaceous rudist bioherms



- High-energy platform margin to low-energy lagoon environments
- Biostromes and patch-reefs
- Reduced role of framebuilding corals: unfavorable environmental conditions

Ancient reefs
Miocene coral reefs, Spain

- Similar to modern-day Scleractinian coral reefs



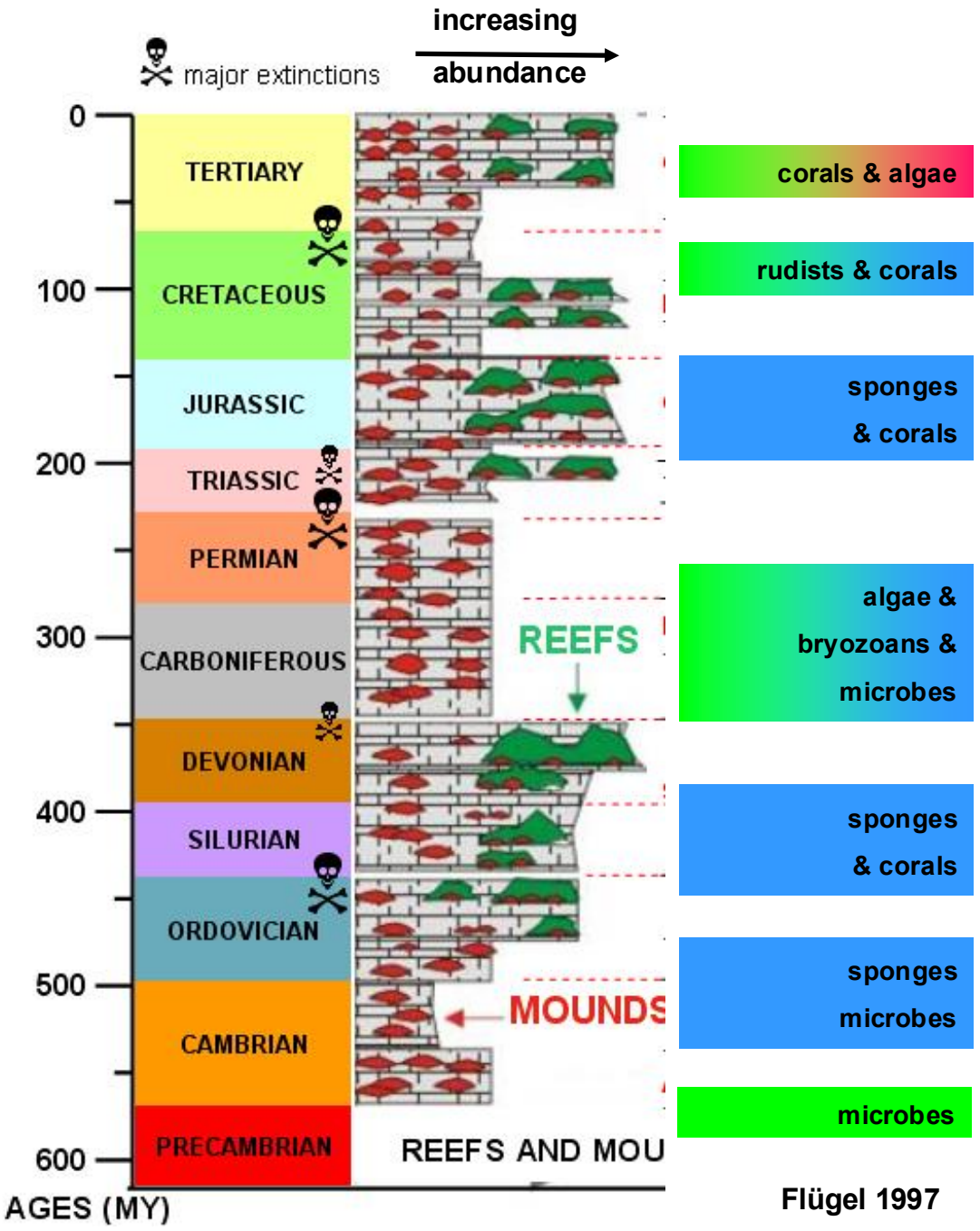
Geopetal structure





Reefs through time
Biological evolution

- The composition of the skeletal components of carbonates has varied through time in response to:
  - Evolution
  - Extinction events
  - Changes in ocean chemistry
  - Changes in continent configuration
- Reefs as organic build-ups are good mirrors of these changes



Flügel 1997

Summary

- **Carbonate build-ups are laterally-restricted structures with organically-mediated growth**
- **Distribution of reef-building organisms strongly controlled by environmental factors (light, wave energy….): preferential growth in shallow water (antecedent topography)**
- **Many ancient reefs had skeletal frameworks building topography and wave-resistant structures (eg, Devonian, Tertiary)**
- **Occurrence of mud mounds in deeper water and when skeletal framebuilders were absent (extinction)**
- **Reading: Tucker (1990) p. 190-227, 401-414**

EART 23202
Reading the Stratigraphic Record

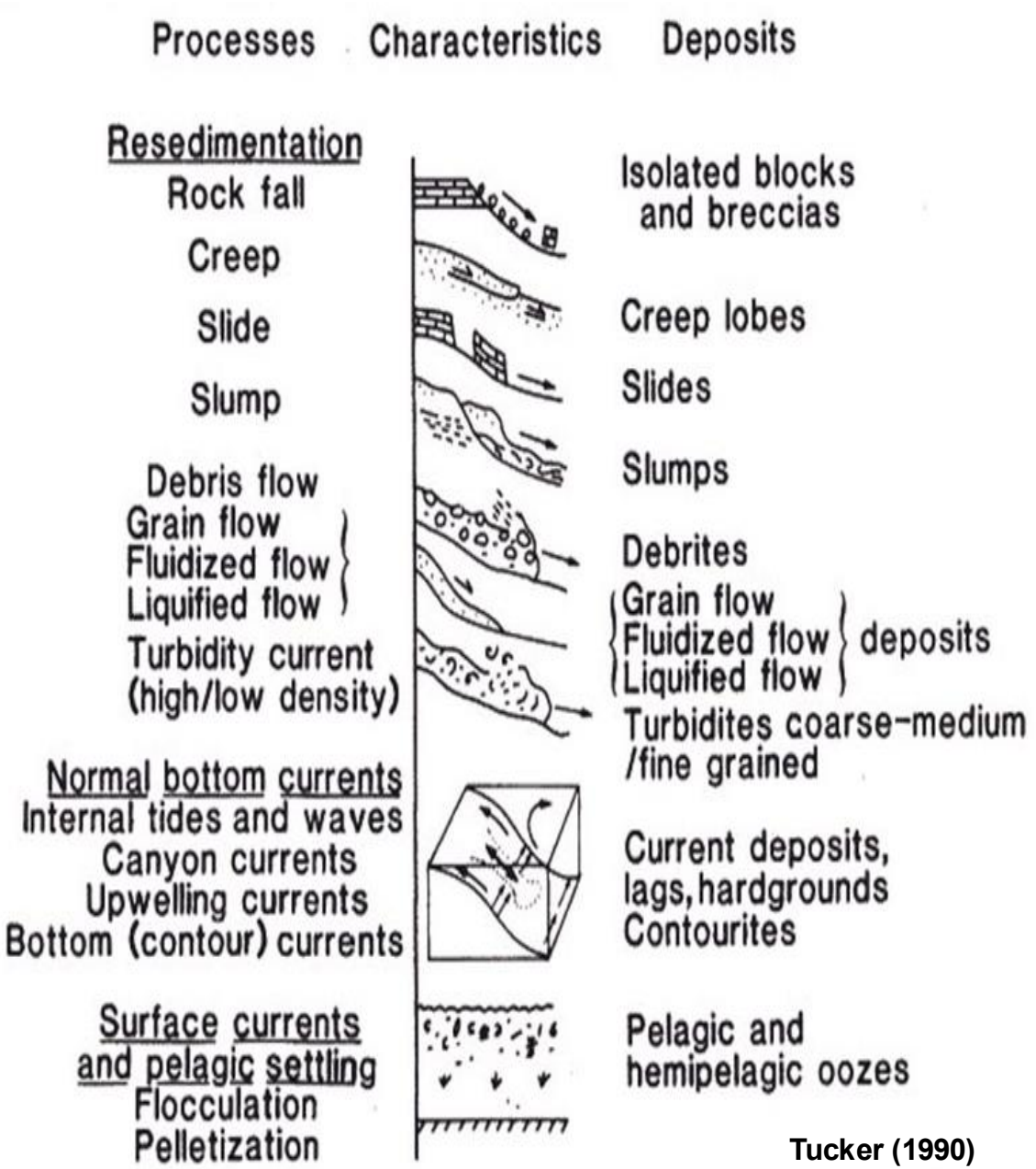


Lecture 5.2: Slope and basin environments

Dr Stefan Schroeder

Slope and Basin depositional environments
Depositional processes

- On ramps and other gentle slopes:
  - Below fair weather wave base - current-/storm-dominated sedimentation
  - Below storm wavebase - fine-grained limestones, siliciclastic shales
- Slopes and basins adjacent to rimmed shelves:
  - Remobilisation
    - Debris/gravity/fluidised flows
    - Calciturbidites
    - Rock fall, slumps
- Deposition from bottom currents
- *in situ* pelagic fall out

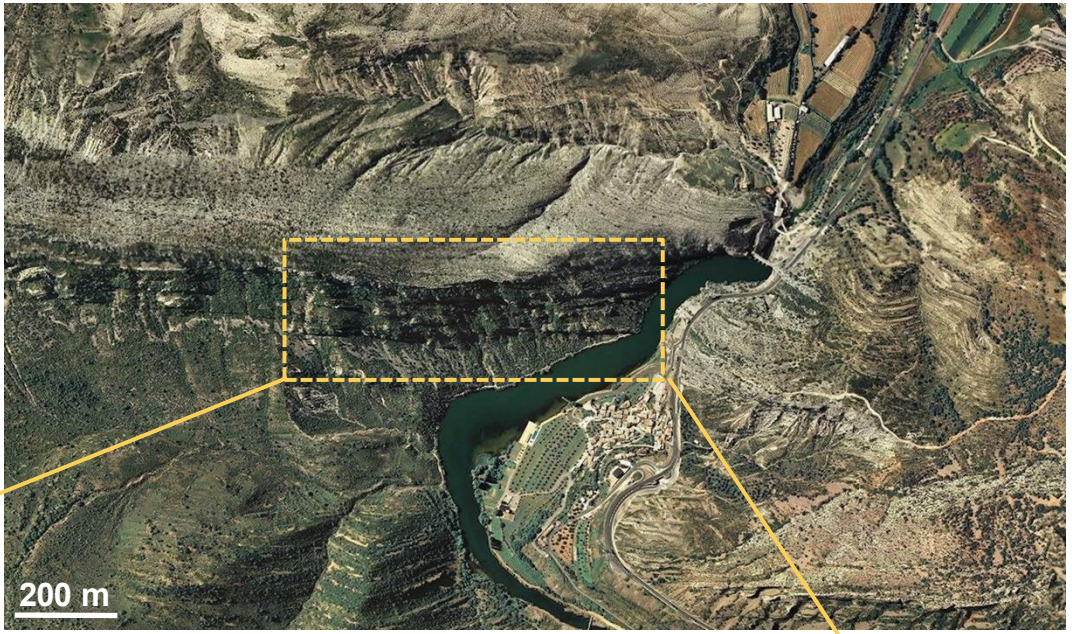


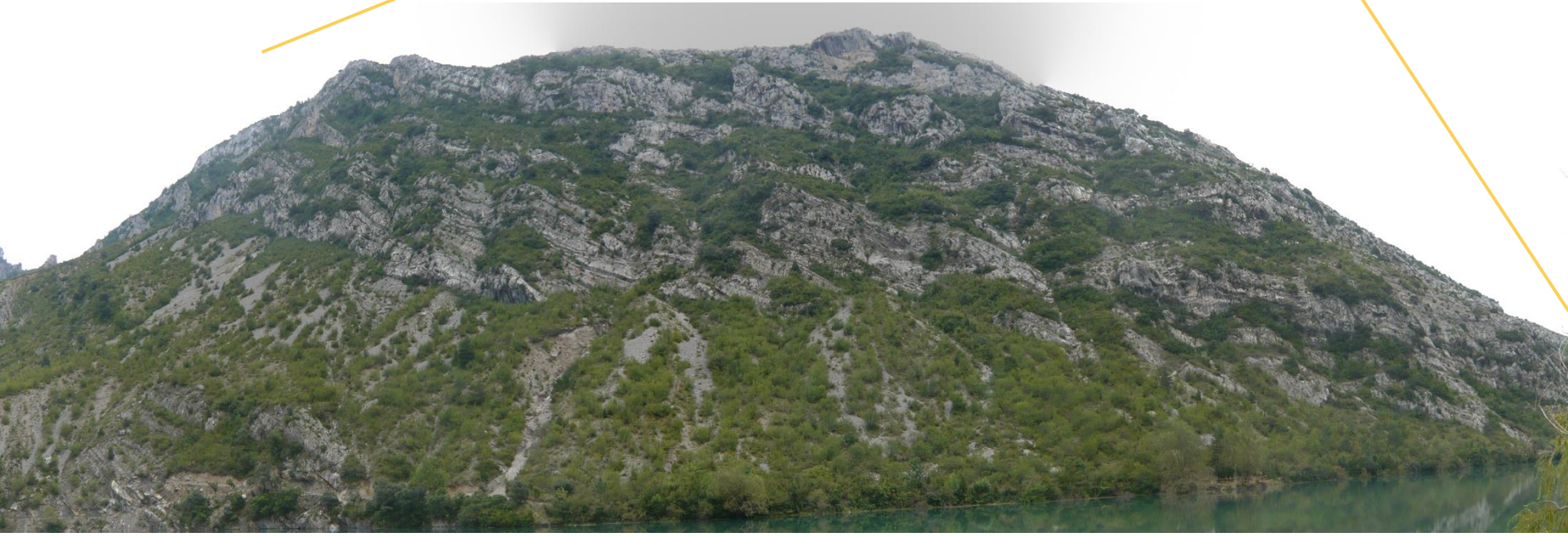
Tucker (1990)

Slope and Basin depositional environments

Depositional processes

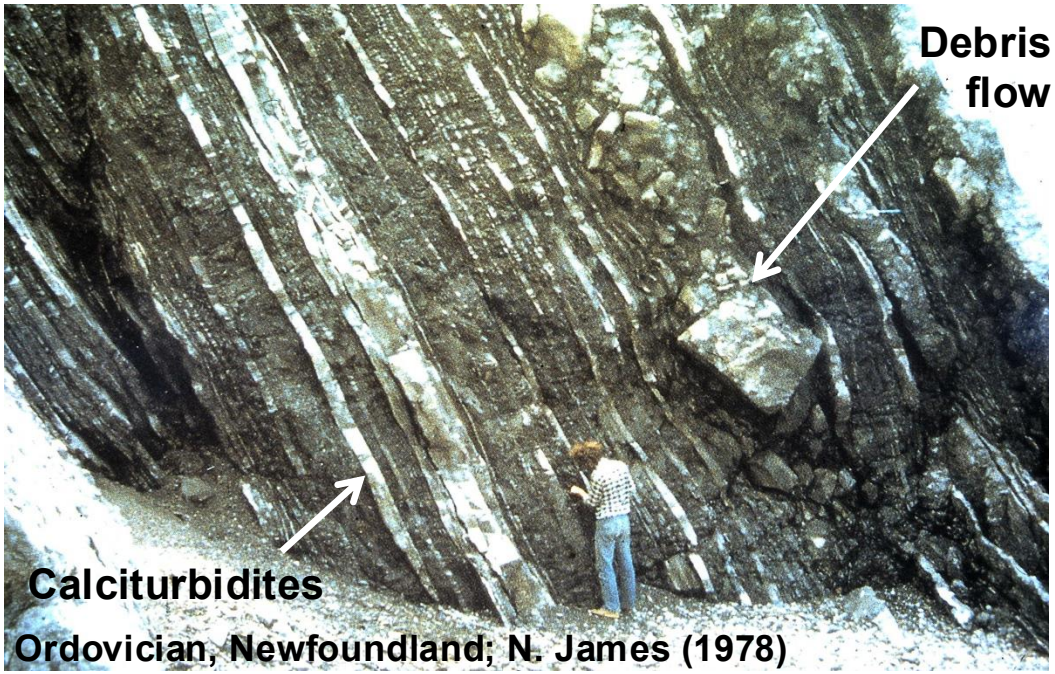
Upper Cretaceous, Pyrenees

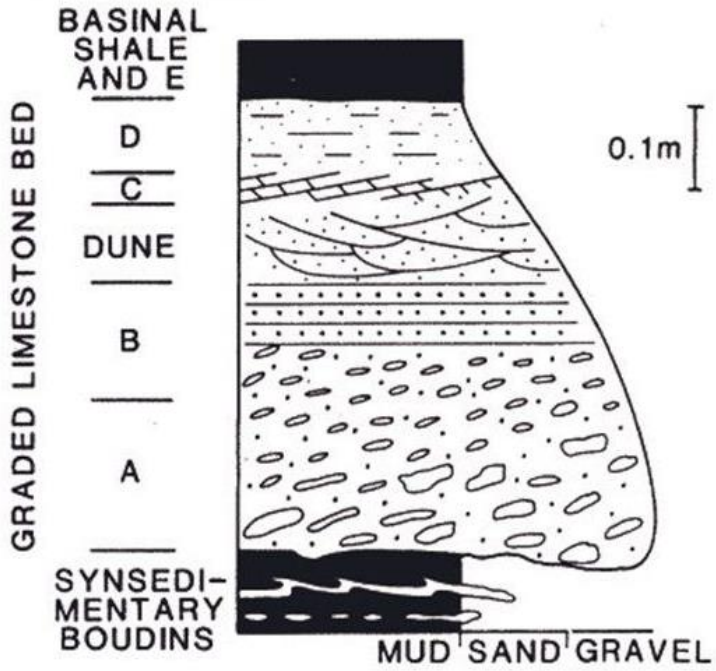




Slope deposition



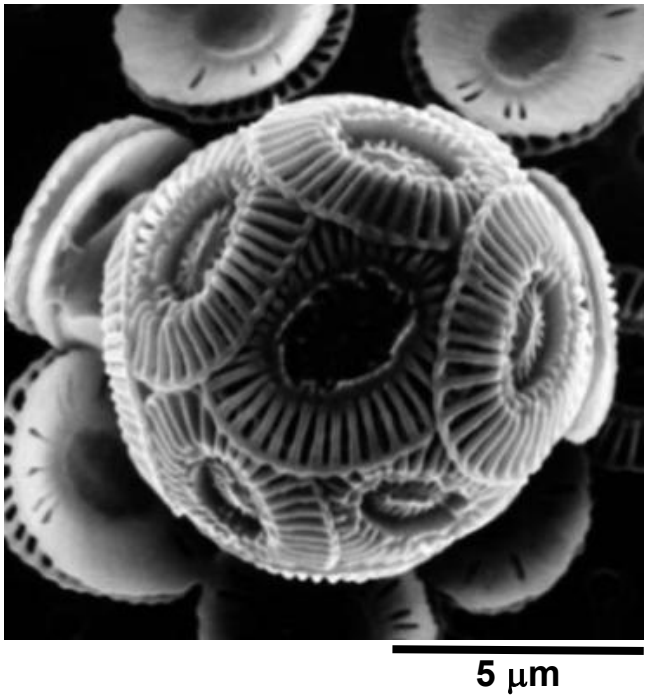


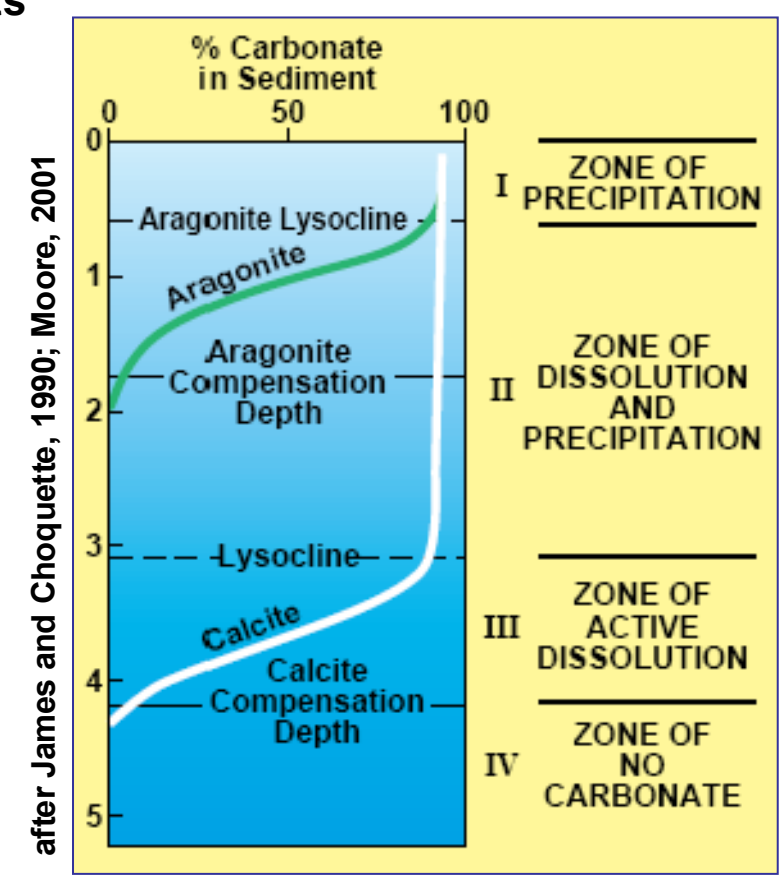


- Turbidites: varying degrees of completeness of Bouma sequence
- Grain flows: well-sorted carbonate sands with reverse grading
- Debris flows: little grading/sorting, rudstones and floatstones

Basin deposition

- Reduced influence of platform margin in deep-water environments
- Settling-out of suspended biogenic material (plankton): pelagic sediments
- Foraminifera, coccolithophorids, diatoms, radiolaria, pteropods…..
- Distribution controlled by productivity and ocean currents







Basin deposition
Pelagic sediments over time

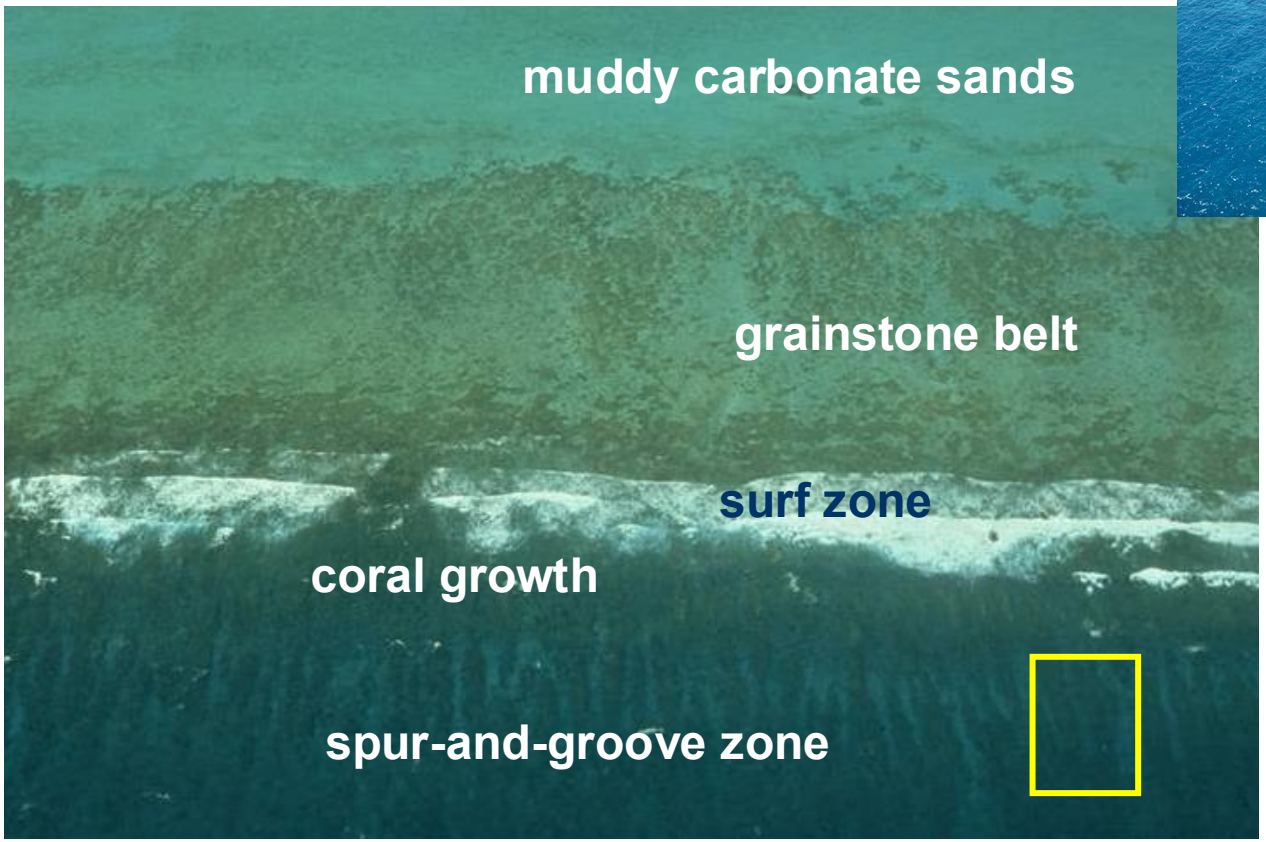
- Nature of plankton changed over time
- Planktonic foraminifera and coccolithophorids only important since Cretaceous
- Triassic-early Jurassic: pelagic bivalves, cephalopods
- Paleozoic: calcareous plankton ~absent
- Radiolaria present since Precambrian



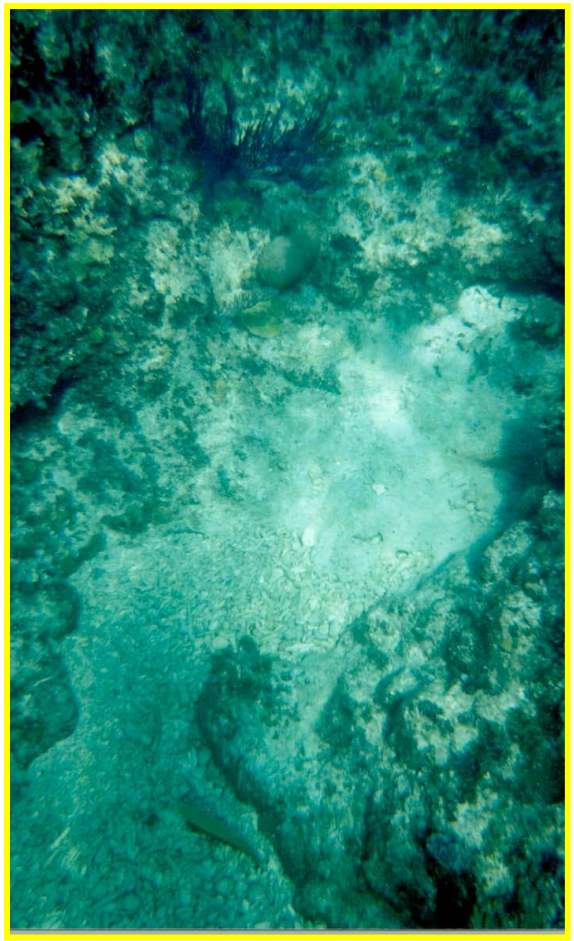
Summary

- Slopes are dominated by (storm) current deposition and gravity-driven deposition. Sediment textures are highly variable; sediment composition reflects source area (reef and/or shoals at platform margin, reworked upper slope).
- Basinal sediments are dominated by pelagic deposition of carbonate mud, commonly mixed or interbedded with siliciclastic shales/mudrocks
- Biogenic component in pelagic sediments is planktonic. Organisms and the resulting sediments varied over time in response to evolution.
- Reading: Tucker (1990), p. 228-236, 238-262, 274-283

Reef facies
Reef front – spur and groove





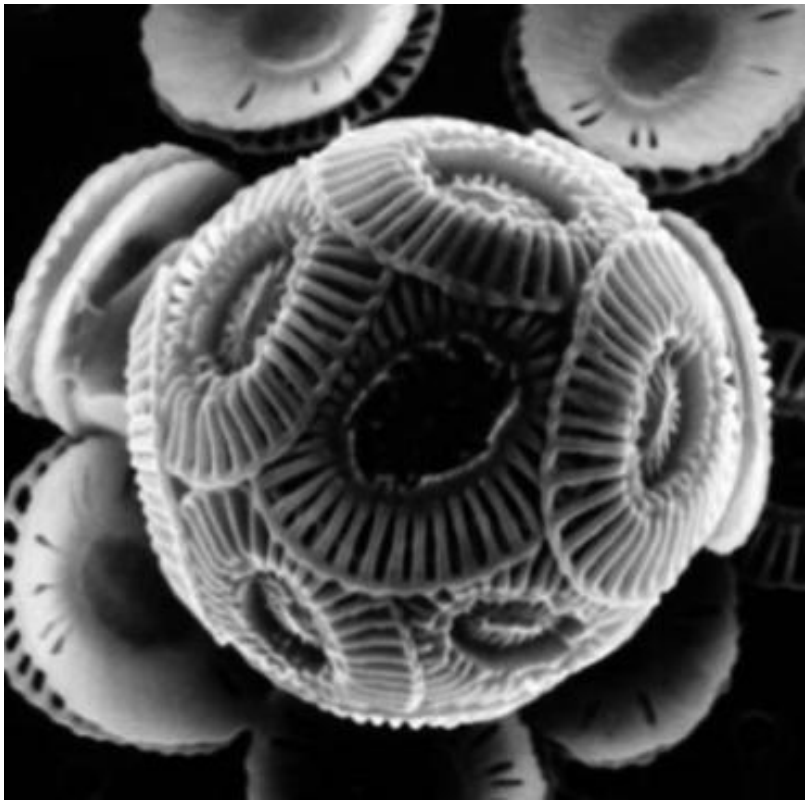


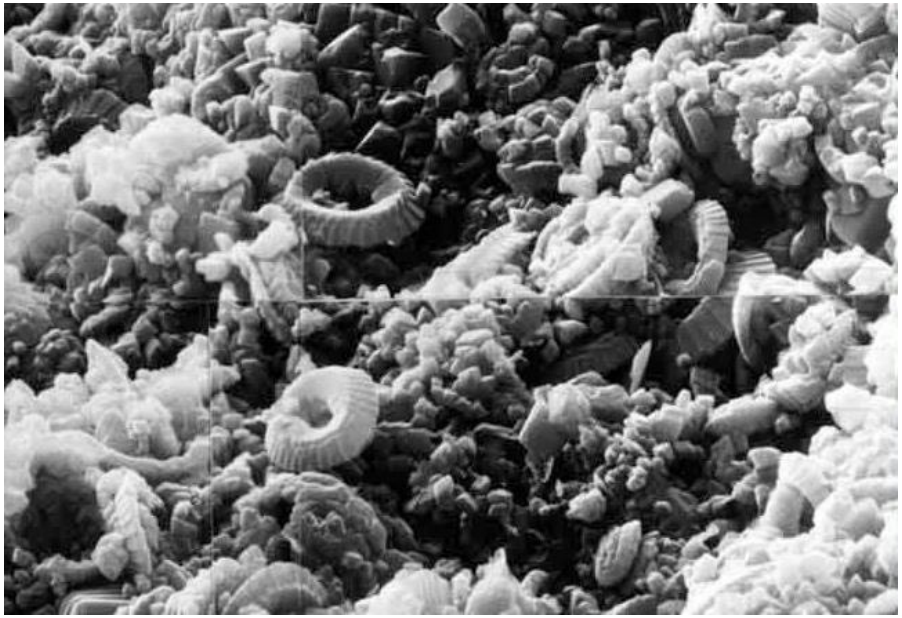
Lighthouse reef, Belize, Central America
Low-altitude oblique aerial photograph showing linear zonation of the Belize Barrier Reef

Cliff Jordan/SEPM

Basin deposition
Planktonic organisms

- Plants with calcite skeletons: Coccolithophorids
- Plants with silica skeletons: eg Diatoms
- Animals with calcite skeletons: planktonic Foraminifera
- Animals with aragonite skeletons: Pteropods (pelagic gastropods)
- Animals with silica skeletons: Radiolaria





Basin deposition
Planktonic organisms

- Plants with calcite skeletons: Coccolithophorids
- Plants with silica skeletons: eg Diatoms
- Animals with calcite skeletons: planktonic Foraminifera
- Animals with aragonite skeletons: Pteropods (pelagic gastropods)
- Animals with silica skeletons: Radiolaria





2 mm

Basin deposition
Planktonic organisms

- Plants with calcite skeletons: Coccolithophorids
- Plants with silica skeletons: eg Diatoms
- Animals with calcite skeletons: planktonic Foraminifera
- Animals with aragonite skeletons: Pteropods (pelagic gastropods)
- Animals with silica skeletons: Radiolaria

